

**DAYTONA STATE COLLEGE
SCHOOL OF ENGINEERING TECHNOLOGY**

**EGN2045
Assignment #06: Matrices**

Solve the following problems.

1) Find the product of the following matrices:

$$\begin{bmatrix} -3 & 6 & -2 \\ 1 & -9 & 4 \\ 10 & 5 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -4 \\ 5 \end{bmatrix}$$

2) For the following matrices, find (a) $A \cdot B$, (b) $B \cdot A$, and (c) $B \cdot B$

$$A = \begin{bmatrix} 4 & -1 \\ -6 & 3 \\ 2 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 6 \\ 5 & 0 \end{bmatrix}$$

3) Find the inverse of the following matrix:

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 1 & 0 & 4 & 1 \\ 2 & 4 & 0 & -2 \\ 4 & 0 & -1 & 2 \end{bmatrix}$$

4) Solve the following system of linear equations:

$$x + 2y + 3z = 30$$

$$2x + 5y + 7z = 71$$

$$3x + 7y + 8z = 87$$

5) Solve the following system of linear equations:

$$6x + 4y + 3z + 2w = 19$$

$$2x - 3y - 5z + w = -2$$

$$x + y - z + 5w = 12$$

$$3x - 2y - 5z + w = -2$$